

Google patents Google Doodle

It has taken Google 10 long years to win a patent on its Google Doodle under 'Systems and methods for enticing users to access a web site'



3D Microscope lens

Two engineers from Ohio State University have designed a 3D lens which enables microscopic objects to be seen in 3D

Feature

SLI Anti Aliasing or Super Anti Aliasing mode:

This mode increases the image quality of the game rather than the performance. It can allow you to crank up your anti-aliasing modes all the way upto 32x.

You also need a bridge connector to combine two cards in SLI or Crossfire with the video coming out of one card, after it has used the processing power of both cards.

Lucid Hydra chip

The Lucid 200 series (comprising three chips) is an application specific integrated circuit (ASIC) built on a 65nm process and consumes under 10W of power thus requiring no heatsink. The ASIC connects to the NorthBridge or directly to the CPU (in case of Sandy Bridge processors) via one x16 upstream port. The downstream port which connects to the PCIe x16 lanes in three combinations – two x16 (for dual GPU), one x16 and two x8 (for triple GPU) and four x8 (for four GPUs). The Hydra chip comprises a 300MHz internal RISC processor having a 64KB instruction cache and 32KB data cache, a DMA engine and has high bandwidth switching capabilities.

Software and Hardware presence

The Hydra engine comes with a driver which rests between the OpenGL or DirectX architecture and the GPU vendor driver. It is compatible with DirectX 9, 10.1 and 11. This allows it to intercept calls from the game or application to DirectX before the information is passed onto the AMD or NVIDIA drivers. This allows the engine to break up the information into jobs called 'tasks'. The definition of task is not fixed and can change based on the information being sent. It can be anything like a special lighting effect, a 3D object, a specific object in the scene, etc. These tasks are sent over the PCI Express bus to the Hydra chip which quickly processes and splits the tasks between the installed GPUs.

Thus, on the chip level it is lodged on the motherboard between the CPU and the card and can communicate via the PCI Express bus and at the driver level it is placed between Windows direct driver interface and the vendor driver. So in a way, it does not com-

GPU Virtualization using Lucid Virtu



Intel's Sandy Bridge processors left many of us impressed with its Quick Sync Video technology which allows you to transcode videos at unprecedented speed. But at the same time Intel's implementation of the QSV on the motherboards irked many enthusiasts. For starters, owners of P67 boards (which allows overclocking the K-series Sandy Bridge processors) had no access to the

processor graphics whereas H67 board users could take advantage of the Intel HD graphics but could not overclock the processor. For those able to use QSV on an H67 board, there was another shortcoming - it will not work if you have discrete card on your motherboard. Although Intel has announced the Z68 chipset which will allow you to overclock your processor and use the Intel HD graphics, the moment you add in a graphics card, you cannot use Intel HD graphics.

Lucid's Virtu is a GPU Virtualization software that works quite similarly to the Lucid Hydra Chip. Virtu allows a user to use an H6x or Z68 chipset along with a discrete graphics card. The video out is taken from the motherboard and not from the graphics card. Just like Hydra, Virtu will intercept API calls and pass it to the GPU based on the application. While gaming, the output will be sent to the card, whereas while video transcoding the Intel HD graphics takes over. Since the tasks are pre-defined, the need to balance the load between Intel HD graphics and the discrete graphics solution does not arise.

municate with the AMD or NVIDIA drivers at the software level or with the GPUs at the hardware level, thus making it driver and GPU agnostic.

After the assigned tasks to the GPUs are completed, they are returned to the interop layer. This layer composites the frames from the different GPUs and then sends it to the GPU which is attached to the display.

The Hydra engine intelligently divides the rendering workload by intercepting the DirectX calls from the game to the graphic cards unlike the present day multi GPU technology where the task is simply broken down based on the type of rendering ie. Alternate frame rendering or Split frame rendering.

As Hydra allows multiple GPU setup of cards having different processing powers, the load has to be divided smartly so that there is no micro-stuttering in the final output. Micro-stutter is a phenomenon that occurs in multi GPU set ups when there is a lag between the output of the two GPUs. It reduces the performance (ie. lesser frame rates). In a set up involving two cards having different core clocks, it is pointless to have both cards working on the same task, as the output of the card with a higher clock will be faster. This is where load balancing comes into play.

The Hydra driver tries to recognise the GPUs in the system and get a rough idea of its processing power. If a GPU takes longer than expected to complete a particular task, then the Hydra engine will send more tasks to the less loaded GPUs. This updation of the allotment of tasks happens as the game is running. It incorporates a feedback mechanism which determines the performance of the installed graphic cards and balances the loads between them accordingly. **[4]**

LIMITATIONS OF LUCID HYDRA

This technology does come with its set of limitations:

1. This will work only with Windows Vista and Windows 7. So XP users will have to upgrade their operating system. Also, if you plan to run AMD and NVIDIA cards then the OS will have to be Windows 7 as that is the only OS which supports multiple graphic drivers running concurrently.
2. Dual GPU products like the Radeon HD 5970 and GeForce GTX 295 are not supported as their internal PCIe switch is not compatible with Hydra 200 chip.
3. The Lucid Hydra technology supports a limited number of games as compared to vendor specific multi GPU set ups. This number varies depending upon the mode of operation of the Hydra chip.